1.如图 4-1 所示电路，$u_s = 50\cos\omega t$ V，$i_s = 2\sin\omega t$ A，其中 ω=2000rad/s。计算稳态电流 i。

$\dot{U}_s = 50\,V$

$\dot{I}_s = 2\angle -90^\circ = -j2A$

$Z_C = -j25\,\Omega$

$Z_L = j\omega L = j80\,\Omega$

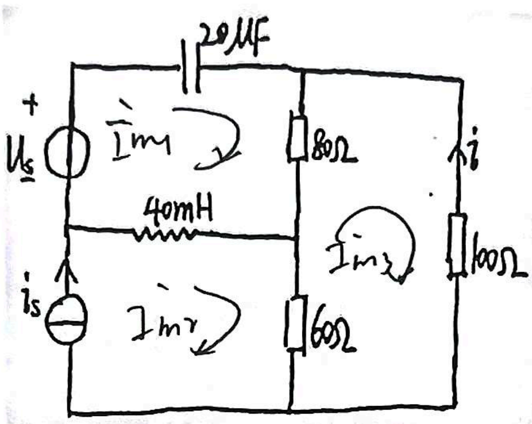


图 4-1

$$\begin{cases} (80+j55)\dot{I}_{m1} - j80\dot{I}_{m2} - 80\dot{I}_{m3} = 50 \\ -80\dot{I}_{m1} - 60\dot{I}_{m2} + 240\dot{I}_{m3} = 0 \\ \dot{I}_{m2} = \dot{I}_s = -j2 \end{cases}$$

$\dot{I} = -\dot{I}_{m3} = 1.38\angle 111.81^\circ$

$i = 1.38\cos(2000t + 111.81^\circ)$

2.如图 4-2 所示电路，Z_L 任意可调。(1) 确定 Z_L 为何值时获得的有功功率最大，并求此最大有功功率。(2) 若将 Z_L 改为电阻负载 R_L，再求最大有功功率。

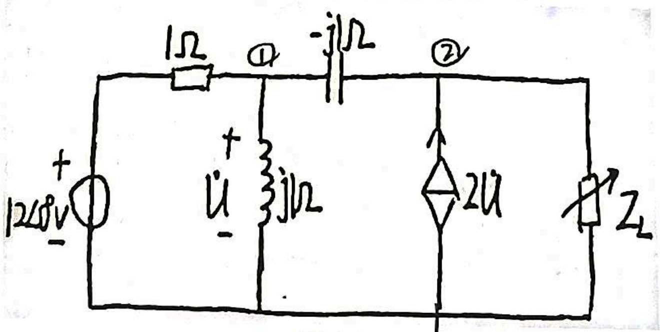


图 4-2

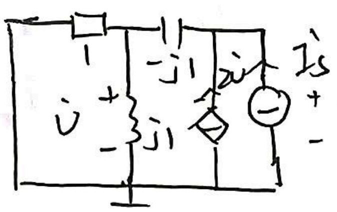
求 $\dot{U}_{oc}$：

$(\frac{1}{1} + \frac{1}{j1} + \frac{1}{-j1})\dot{U}_{n1} - \frac{1}{-j1}\dot{U}_{n2} = 12$

$-\frac{1}{-j1}\dot{U}_{n1} + (\frac{1}{-j1})\dot{U}_{n2} = 2\dot{U}$

$\dot{U}_{oc} = (6+j18)\,V$

求 Z_i：外加电流源 $\dot{I}_s$ 独立源置零



$\frac{\dot{U}_s}{(1\|j1)-j1} = 2\dot{U} + \dot{I}_s$

$\dot{U} = \frac{1\|j1}{(1\|j1)-j1}\dot{U}_s$

$Z_i = \frac{\dot{U}_s}{\dot{I}_s} = (\frac{1}{2} + j\frac{1}{2})\,\Omega$

(1) $Z_L = 0.5 - j0.5\,\Omega$ 时 $P_{max} = 180W$

(2) 当 $R_L = \frac{\sqrt{2}}{2}\,\Omega$ 时 $P_{max} = 149.12W$

3.如图 4-3 所示电路，$u_s = 100\sqrt{2}\sin 10t$ V，Z_L 任意可调。确定 Z_L 为何值时获得的有功功率最大，并求此最大有功功率。

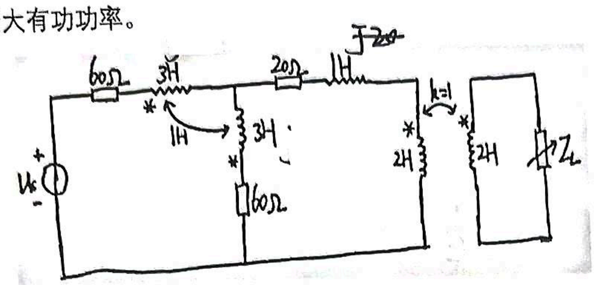
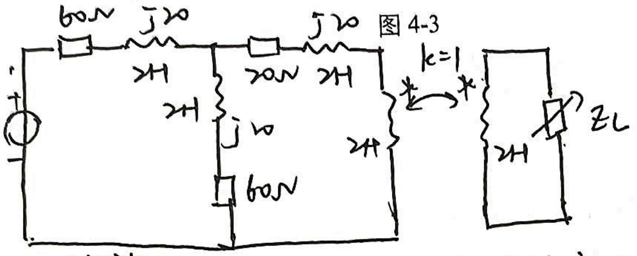


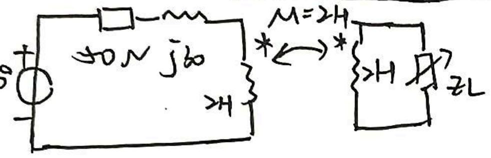
图 4-3

等效电路：



$k=\frac{M}{\sqrt{L_1L_2}}=1=\frac{M}{\sqrt{2\times2}}$

$\therefore M=2H$



将左侧电路等效到右侧

$Z_{eq}=Z_r+j\omega L_2=\frac{(\omega M)^2}{总阻抗}+j\omega L_2=\frac{(20)^2}{50+j30+j20}+j20$

$=j16+4$

$\therefore Z_L=4-j16$时 P_{max}. 再将右侧电路等效到左侧 $Z_{eq}'=\frac{(\omega M)^2}{4-j16+j20}+j\omega L_1=50-j30$

$\dot U_{oc}=\frac{1}{2}\dot U_s=50V$

$P=\frac{50^2}{4\times50}=12.5W$

4.如图 4-4 所示电路中的容抗 X_C 和匝数比 n 分别为多少时可以使得 Z_F 获得最大功率，并求此最大功率，设 $Z_F=(80+j60)\Omega$

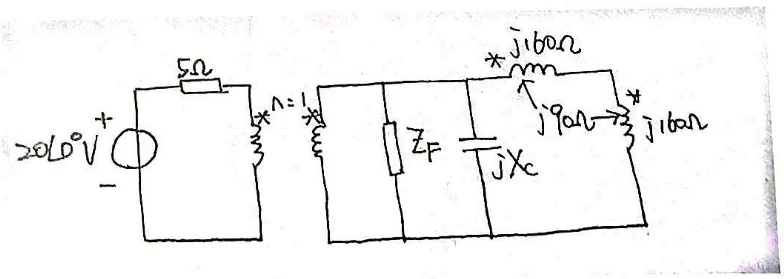


图 4-4

右侧 $Z_L=j\omega L_1+j\omega L_2+2j\omega M=j(160+160+2\times90)=j500\Omega$

副边导纳 $Y=\frac{1}{80+j60}+\frac{1}{jX_C}+\frac{1}{j500}=0.008+j(-\frac{1}{X_C}-0.008)$

等效到原边 $Z_1=n^2Z=\frac{n^2}{Y}$　　$-\frac{1}{X_C}-0.008=0$时　$X_C=-125\Omega$　P_{max}

$Z_1=n^2\cdot125=5\Omega$时　$n=0.2$

$\therefore n=0.2$　$X_C=-125\Omega$　$P_{max}=\frac{20^2}{4\times5}=20W$

5.用节点法列写图 4-5 中所示电路的相量方程

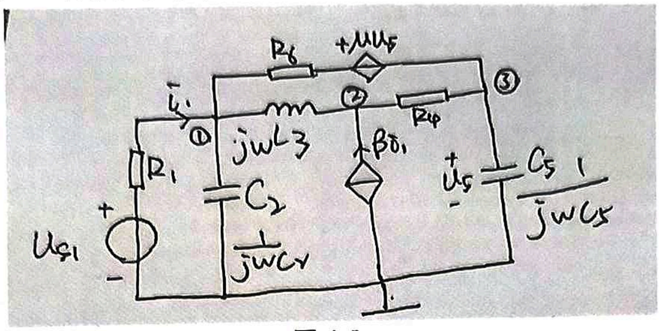


图 4-5

$$\begin{cases} \left(\frac{1}{R_1}+\frac{1}{R_6}+\frac{1}{j\omega L_3}+j\omega C_2\right)\dot{U}_{n1}-\frac{1}{j\omega L_3}\dot{U}_{n2}-\frac{1}{R_6}\dot{U}_{n3}=\frac{\mu\dot{U}_5}{R_6}+\frac{\dot{U}_{s1}}{R_1} \\ -\frac{1}{j\omega L_3}\dot{U}_{n1}+\left(\frac{1}{j\omega L_3}+\frac{1}{R_4}\right)\dot{U}_{n2}-\frac{1}{R_4}\dot{U}_{n3}=\beta\dot{I}_1 \\ -\frac{1}{R_6}\dot{U}_{n1}-\frac{1}{R_4}\dot{U}_{n2}+\left(\frac{1}{R_4}+\frac{1}{R_6}+j\omega C_5\right)\dot{U}_{n3}=-\frac{\mu\dot{U}_5}{R_6} \\ \dot{U}_5=\dot{U}_{n3} \\ \dot{I}_1=\frac{\dot{U}_{s1}-\dot{U}_{n1}}{R_1} \end{cases}$$

6.求出图 4-6 中等效阻抗 Z_i $Z_i=(11.92-j4.62)\Omega$

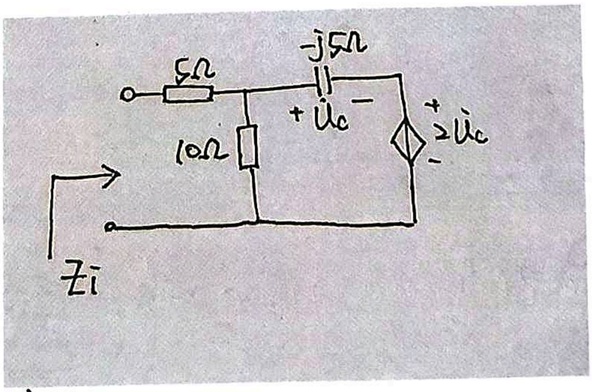


图 4-6

$$\dot{I}=\frac{3\dot{U}_c}{10}+\frac{\dot{U}_c}{-j5}=\dot{U}_c\left(\frac{3+j2}{10}\right)$$

$$\dot{U}=\dot{I}\cdot 5+3\dot{U}_c=\frac{9+j2}{2}\dot{U}_c$$

$$Z_i=\frac{5(9+j2)}{3+j2}=(11.92-j4.62)\Omega$$